Adrian Erlikhman

(213) 330-6884 | erlikhman.adrian@gmail.com | adrianerlikhman.is-a.dev | LinkedIn | GitHub

Education

Los Angeles Center for Enriched Studies (LACES) Class of 2027

LAUSD magnet school | Los Angeles, CA

- GPA 5.2 weighted / 4.0 unweighted | SAT 1520 (Math 770, EBRW 750) | 2× AP Scholar with Distinction
- 17 AP courses, 11 completed and 6 this year, across computer science, mathematics and the sciences.

Los Angeles Southwest College

Dual enrollment | Los Angeles Community College District

- Linear Algebra (MATH 270) | Programming in Python (CS 119) | Python Programming for Data Science and Machine Learning (CS 121, planned)

Research Interests

Evaluating language models and the tests used to rank them: authorship attribution and stylometry, LLM judges, and how reward models and judges are chosen. Also the robustness of standard quantitative methods, in portfolio construction and in composite indices.

Publications

Conference and workshop papers

[1] **A. Erlikhman**. "Curated to Win: Certifying Reward-Model and Judge Selection Against Evaluation-Set Curation." *First Workshop on Autonomous Research and Recursive Self-Improvement (ARRSI), AACL-IJCNLP 2026*, Nov 2026. **Accepted.** Sole author. PDF Code

- Proposed a curation certificate for choosing a reward model or LLM judge by validation accuracy: how much of the set could be removed before the winner changes. In benchmark weight it is at least the top-two gap; counted in items it is exact.
- Removing 12 particular items of RewardBench's 2,985 makes the runner-up the winner (11 under its own scoring code). On 200-item samples, picks with a top-two gap above 2 points match the full-benchmark winner 87% of the time (43% otherwise).

[2] **A. Erlikhman**, M. Tarekegn, P. Juang. "Interpretable Multilingual Attribution of Frontier LLMs." *IEEE BigData 2026 High School Symposium*, Phoenix, AZ, Dec 2026. **Accepted; Best Paper Finalist.** Co-first author. PDF

- 840 essays from five frontier models in seven languages, English to Hindi, attributed with 21 interpretable features from Universal Dependencies parses: 59–72% accuracy in every language against a 20% chance floor, with no decline detected as a language's resources fall.
- The features transfer between languages better than character n-grams (49% against 41%) and hold at 57–70% on machine-translated text; across 4,195 judge calls, four of the five models attribute authorship at chance.

Journal articles

[3] **A. Erlikhman**, R. Erlikhman. "Does Portfolio Optimization Beat Splitting Evenly?" *Oxford Journal of Student Scholarship* 12(7):510–529, Oct 2026. **Published.** Co-first author. PDF

- Equal weights beat Markowitz in 74% of 250 calibrated simulations (Sharpe 0.70 against 0.51). The loss traces to estimating expected returns, and shrinkage closes it only with roughly two decades of stationary data.

[4] **A. Erlikhman**, R. Erlikhman. "How Robust Are the Legatum Prosperity Index Rankings? A Monte Carlo and Sensitivity Analysis of Weighting, Normalization, and Aggregation." *Journal of High School Science*. **In revision.** Co-first author. PDF

- Reproduced every published score for 167 countries, then ran 10,000 draws under each of three weight priors and a 131,072-run Sobol' design over weighting, normalization and aggregation.
- Derived rank leverage, $c^2(1-\rho^2)$, which tracks Sobol' total effects (Spearman 0.91–0.99) at a fraction of the compute. The ranking holds as a whole (median Spearman ≥ 0.9899) but not mid-table, where the median 90% rank interval spans 36 places.

Preprints

[5] **A. Erlikhman**, M. Tarekegn, P. Juang. "Whose Line Is It Anyway? Auditing Self- and Other-Recognition in LLM Judges." Preprint, arXiv forthcoming. Co-first author. Code and data

- Introduced a peer baseline that scores each LLM judge's self-attribution against how often the other judges name it on the same text, separating self-recognition from text any model can identify.

- Nine judges, 1,270 texts, over 19,000 judgments: the usual test flags 15 of 37 cases, 8 survive a false-alarm control, 3 beat the pooled peers and none beats its strongest single peer. An 18-feature stylometric classifier finds the true author 86.3% of the time, above every judge.

Presentations

"Interpretable Multilingual Attribution of Frontier LLMs." *IEEE BigData 2026 High School Symposium*, Phoenix, AZ. Best Paper Finalist. Dec 2026

Student panel on AI (invited), aiEDU pre-conference session "Transform Your School's AI Readiness." *NCSS 2026, National Consortium of STEM Schools*, San Diego, CA. Dec 2026

"Curated to Win." *ARRSI Workshop, AACL-IJCNLP 2026*. Virtual. Nov 2026

Invited guest, aiEDU podcast, hosted by aiEDU CEO Alex Kotran.

The AIML-LI curriculum. *LAUSD Innovation Expo*, Los Angeles, alongside Google, Microsoft and Canva.

Honors & Awards

Best Paper Finalist, IEEE BigData 2026 HS Symposium (one of 12; top 16 of 130 accepted papers) 2026

National Merit Semifinalist: about 16,000 of 1.3M+ entrants 2026

1st place, Council District 9 Champion (U18), VISION HACK: South LA, for Ledger 2026

1st place, Finance track, ACM TrojanHacks, for Monte 2026

Winner, Agent Behaviour Monitoring track, SEPHacks, judged by Lemma, for Hivemind 2026

1st place, Decode the Ocean Hackathon (Lovable × United Nations), for eDNAtlas 2026

3rd place internationally, Code for Transportation (Young Coders' Sphere), for Safe Routes to School 2026

Finalist, top 8 of 45 teams, Citadel High School Terminal 2026

Admitted, Y Combinator Startup School, as AIML-LI's founder 2026

JFEDLA Teen Innovation Grant, for SafeJew 2024–25

2× AP Scholar with Distinction

USA Fencing: 3× All-American and 3× All-Academic First Team

SIFMA Stock Market Game: top 5% statewide, as a solo entrant

Experience

Firstness 2026 – Present

Software Engineering & ML Intern | firstness.ai | reporting directly to the founder

- Work directly under the founder of Firstness, an AI research engine that grows a research question into a map of branching hypotheses, writes each answer from the peer-reviewed literature with a citation on every sentence, and checks a new hypothesis against published prior work.
- Refining how the platform connects into the research stack: the literature databases and reference managers researchers already work in, so an answer arrives inside the tools they use rather than in one more tab.
- Building the demos that walk a new user through what the engine does, why every claim carries a citation, and where each answer can be checked against its source.

Kiddom 2026 – Present

Machine Learning Internship & Mentorship | kiddom.co | with Director of Data Science Flora Xu

- Designing machine-learning and NLP approaches to teacher-facing problems in K–12 assessment: which problems are worth modeling at all, what student data a school can realistically share, and what result would count as evidence rather than a demo.
- Built **Why Wrong** against one of them. It reads a class's answer sheets, names the misconception behind each wrong answer (naive Bayes over answer patterns, then clustering on the residual errors), and groups the class by shared mistake. F1 0.87 on planted misconceptions. live code
- Designing its classroom evaluation with an LAUSD teacher: teacher review of the item bank and the misconception labels, a pilot on de-identified responses from a single class, then a written comparison of where the tool and the teacher agree and where they do not.

Fjor Venture Capital 2024 – 2026

Investment & Software Engineering Intern | $15M early-stage fund

- Built classifier and scoring models to screen 134 startups in Y Combinator's Summer 2025 batch, and led the screen with Managing Director Tatiana Janko: sourcing, first-pass screening and the competitive landscape behind each shortlisted company.
- Ran diligence on 20+ early-stage AI startups across enterprise SaaS, fintech and deep tech, and wrote 10+ full investment memoranda for partner-level decisions.
- Built the fund's internal software tools, and drafted its monthly LP updates and investor-facing materials.

Alliance for SoCal Innovation 2024 – 2026

Venture & Data Strategy Intern | nonprofit | SoCal startup ecosystem

- Built a Python/SQL investor-matching engine over 10K contacts at 250+ VC firms, angel syndicates and accelerators, matching startups to likely investors from investment histories and company characteristics; deduplication and standardization cut founder-to-investor lookups from days to under a minute.
- Led the backend redesign of the venture pipeline with the Chief Innovation Officer; led outreach to 80+ founders and mapped the SoCal deep-tech ecosystem.

Advanced Mathematics Research 2022 – Present

induction, number theory, combinatorics and game theory, with PhD candidate Kirill Kovalenko

Selected Projects

AIML-LI (AI/ML Literacy Initiative) 2023 – Present

Co-Founder & Lead | aiml-initiative.org

- Co-wrote a free AI-literacy course any teacher can run in twelve 30-minute sessions on the Chromebooks a class already has, plus a 12-week version in six units, with Colab notebooks, instructor guides, student packets and assessments, all open-source. Taught in LACES AP Computer Science, spring 2026.
- Took it to LAUSD's Ed Tech & Innovation office (Sept 2026), which is carrying it to the district's Chief Academic and Chief Information Officers toward a standing AI-literacy elective. Advised by Philo Juang (Google DeepMind) and Amir Zohrenejad (Data Herald, YC W21); mentored by aiEDU.
- Organizing the LA Student AI Summit & Hackathon for 500+ LA public-school students, with keynotes from Google DeepMind, Microsoft, Google and Snap.

SafeJew 2024 – Present

Co-Founder | safejew.org | JFEDLA Teen Innovation Grant

- A live map of antisemitic incidents across Greater LA: 800 incidents, 100 of them individually sourced with a citable link on every pin, plus 594 on 33 university campuses. A dependency-free LA gazetteer geocodes the places named in news prose, so a daily GitHub Action keeps it current.
- Distributed through the Jewish Federation of Los Angeles' Community Security Initiative, which funded the build.

Safe Routes to School 2026

Co-Creator | safe-routes-la.github.io

- Scores 431,599 LA street blocks for street-crime risk from 85,634 filtered LAPD incidents, then runs A* over them in the browser across the 668 LAUSD schools. Riding the bus cuts a route's modeled exposure by up to 80% while getting there faster.

Ledger Sep 2026

VISION HACK: South LA | Hacker Fund × City of LA Vision Lab

- Led the twelve-hour build across the vision pipeline, rule engine and frontend, and wrote the two-pass vision pipeline, the dashboard and renewal calendar, and the static fallback. Ledger puts the 18 programs a corner store answers to on one renewal calendar, and scores a photo of a wholesale order record against the SNAP stocking rule. live code

Hivemind Oct 2026

SEPHacks 2026 | with Michael Tarekegn

- Shared memory for a team's AI coding agents: an engineer's correction becomes a lesson, the team lead approves it, and every agent loads it at its next session. Built the website, the web app and the terminal app's design, and made the launch film; Michael built the engine and eval. Claude Code passed 36 of 36 eval runs with it, 14 of 36 without. live devpost

Monte Oct 2026

ACM TrojanHacks 2026 | with Michael Tarekegn

- Picks the credit card that pays most at each purchase, never one carrying a balance. Built the interface, the tap-to-pay demo and the brand, and made the launch film; Michael built the backend, the Monte Carlo and the tests. live devpost

dunderscore 2026

Citadel High School Terminal | team of three

- A Python tower-defense strategy that reads the board, prices every wall, turret and attack lane, and adapts to the opponent's last layout; a new version shipped after each round's replay review.

eDNAtlas 2026

Decode the Ocean Hackathon | Lovable × United Nations

- A live map that turns environmental DNA sampling into a plain-language health score for coastal sites, with the raw evidence and Darwin Core downloads behind every score. live code

Leadership & Service

Jewish Student Union 2023 – Present

President | 50+ members

- Run weekly programming with NCSY; organized a school Holocaust Remembrance assembly with a survivor and StandWithUs.

STEMsters 2022 – Present

Co-Founder & President | 30+ volunteers | 4 elementary schools

- Place LACES volunteers in four elementary schools to teach hands-on coding, robotics and engineering.

LA International Fencing Center

Coach

- Coach the club's younger épéeists, including my brother, through their first competitive season.

Athletics

Competitive Fencing, Épée

USA Fencing A26, Team USA | LA International Fencing Center

- U.S. No. 49 Junior (U20) and No. 78 Senior; peak No. 17 Cadet (U17); No. 1 in Region 4. Represented Team USA at the 2026 Cadet Pan-American Championships in Bogotá (6th). In active recruitment with MIT.
- Top 32 Division I, National Championships | Top 8 Cadet, SJCC | Top 16 Cadet, October North American Cup | 4× Junior regional champion

Skills

Code: Python, TypeScript/JavaScript, Java, SQL, LaTeX; Next.js, React, Supabase/PostgreSQL, GitHub Actions, Git

ML & statistics: PyTorch, scikit-learn, TensorFlow/Keras, pandas, NumPy, SHAP, Stanza, FinBERT; LLM evaluation, stylometry, Monte Carlo and Sobol' sensitivity analysis, hypothesis testing, time series, HMMs, A* search

Certifications: Certiport IT Specialist in Software Development, Cybersecurity, Python, Data Analytics, AI and HTML/CSS; Deep Learning Specialization (Coursera); NVIDIA Deep Learning Fundamentals; Kaggle × Google Generative AI Intensive

Music: MTAC piano, Level 9 candidate, five state-honors exams

Languages: English, Russian (fluent), Spanish (intermediate)